

Midterm Mastery Part III

Ticket #1

1. Bug Location

server.js line 35

2. Explanation

(1) Root cause

The server port was determined only by the environment variable APP_PORT.

If APP_PORT was not defined, the value of port became undefined.

In Node.js, calling `app.listen(undefined)` does not throw an error. Instead, the server binds to a random ephemeral port assigned by the operating system. This makes the API unreachable from the expected port (9000), causing the service to appear down.

(2) Fix

Added a fallback value so the server defaults to port 9000 when APP_PORT is not set.

3. Code Diff

```
humming-bird-midterm $ git diff server.js
diff --git a/server.js b/server.js
index 454790a..46ce716 100644
--- a/server.js
+++ b/server.js
@@ -32,7 +32,7 @@ app.get('/health', (req, res) => {

  app.use('/v1/media', mediaRoutes);

-const port = process.env.APP_PORT;
+const port = process.env.APP_PORT || 9000;
```

Ticket #2

1. Reproduce the Bug

```
humming-bird-midterm $ curl -X POST "$ALB/v1/media/upload?width=800" -F "file=@sample.jpg"
{"mediaId":"b62ee571-4335-409a-a83e-32228530220b"}humming-bird-midterm $
humming-bird-midterm $ export MID=b62ee571-4335-409a-a83e-32228530220b
humming-bird-midterm $ curl "$ALB/v1/media/$MID"
{"mediaId":"b62ee571-4335-409a-a83e-32228530220b", "size":32553, "name":"sample.jpg", "mimetype":"image/jpeg", "status":"PENDING"}humming-bird-midterm $
```

2. Bug Location

clients/dynamodb.js – getMedia()

line 78 – 84

3. Explanation

(1) Root cause

The createMedia function saved the width attribute to DynamoDB, but the getMedia function did not include it when constructing the API response.

As a result, the metadata returned by GET /v1/media/:id did not contain the width field even though it was stored in the database.

(2) Fix

Added the missing field to the returned object.

4. Code Diff

```
{ "mediaId": "b62ee571-4335-409a-a83e-32228530220b", "size": 32553, "name": "sample.jpg", "mimetype": "image/jpeg", "status": "PENDING" } humming-bird-midterm $ git diff clients/dynamodb.js
diff --git a/clients/dynamodb.js b/clients/dynamodb.js
index 700494c..d25d01f 100644
--- a/clients/dynamodb.js
+++ b/clients/dynamodb.js
@@ -81,6 +81,7 @@ const getMedia = async (mediaId) => {
   name: Item.name.S,
   mimetype: Item.mimetype.S,
   status: Item.status.S,
+  width: Number(Item.width.N),
 };
 } catch (error) {
   logger.error(error);
 }
```

5. Verify the Fix Works

```
clients $ curl "$ALB/v1/media/$MID"
{"mediaId":"df441c58-50b2-4a10-a361-1451b0170662","size":32553,"name":"sample.jpg","mimetype":"image/jpeg","status":"PENDING","width":800}clients $
```

Ticket #3

1. Reproduce the Bug

```
humming-bird-midterm $ curl -i "$ALB/v1/media/$MID/download"
HTTP/1.1 202 Accepted
Date: Mon, 09 Mar 2026 06:45:33 GMT
Content-Type: application/json; charset=utf-8
Content-Length: 43
Connection: keep-alive
X-Powered-By: Express
Retry-After: 60
Location: hummingbird-production-alb-1041159599.us-west-2.elb.amazonaws.com/v1/media/59ce809c-d422-40b2-971f-8c11fc59bc2b/status
ETag: W/"2b-DVJ8jHebx01CN9tAyyv9tXcsioNc"
```

2. Bug Location

controllers/media.js – downloadController

line 111

3. Explanation

(1) Root cause

The Location header was constructed using req.hostname, which only returns the hostname and does not include the URL scheme (http://). As a result, the returned value was not a valid absolute URL.

(2) Fix

Rebuilt the Location header using req.get('host') and explicitly adding the http:// scheme.

4. Code Diff

```
humming-bird-midterm $ git diff controllers/media.js
diff --git a/controllers/media.js b/controllers/media.js
index 6bb1c50..a148f48 100644
--- a/controllers/media.js
+++ b/controllers/media.js
@@ -108,7 +108,7 @@ const downloadController = async (req, res) => {
  if (media.status !== MEDIA_STATUS.PROCESSING) {
    const SIXTY_SECONDS = 60;
    res.set('Retry-After', SIXTY_SECONDS);
-   res.set('Location', `${req.hostname}/v1/media/${mediaId}/status`);
+   res.set('Location', `http://${req.get('host')}/v1/media/${mediaId}/status`);
  }
  logger.info(
    { mediaId, currentStatus: media.status },
    'Media not ready for download, sending 202'
```

5. Verify the Fix Works

```
humming-bird-midterm $ curl -i "$ALB/v1/media/$MID/download"
HTTP/1.1 202 Accepted
Date: Mon, 09 Mar 2026 07:03:26 GMT
Content-Type: application/json; charset=utf-8
Content-Length: 43
Connection: keep-alive
X-Powered-By: Express
Retry-After: 60
Location: http://hummingbird-production-alb-1041159599.us-west-2.elb.amazonaws.com/v1/media/776b6080-fbf8-41b9-82f8-008ff46346a0/status
ETag: W/"2b-DVJ8jHebx01CN9tAyy9tXcsioNc"
```

Ticket #4

1. Reproduce the Bug

```
humming-bird-midterm $ curl -i "$ALB/v1/media/$MID/download"
{"status":"COMPLETE"}humming-bird-midterm $ curl -i "$ALB/v1/media/$MID/download"
HTTP/1.1 202 Accepted
Date: Mon, 09 Mar 2026 07:38:27 GMT
Content-Type: application/json; charset=utf-8
Content-Length: 43
Connection: keep-alive
X-Powered-By: Express
Retry-After: 60
Location: http://hummingbird-production-alb-1041159599.us-west-2.elb.amazonaws.com/v1/media/0af490bc-8488-45b5-b137-84cefd80f86/status
ETag: W/"2b-DVJ8jHebx01CN9tAyy9tXcsioNc"
```

```
2026-03-09T07:38:27.980000-00:00 api/0831ede5f8a3492a8d1c8b63238a2e {"level":"info","message":{"currentStatus":"COMPLETE","mediaId":"0af490bc-8488-45b5-b137-84cefd80f86"},"service":"hummingbird","timestamp":"2026-03-09T07:38:27.980Z"}
2026-03-09T07:38:27.981000-00:00 api/0831ede5f8a3492a8d1c8b63238a2e {"level":"info","message":{"duration":38,"method":"GET","statusCode":202,"url":"/v1/media/0af490bc-8488-45b5-b137-84cefd80f86/download"},"service":"hummingbird","timestamp":"2026-03-09T07:38:27.981Z"}
```

2. Bug Location

controllers/media.js – downloadController
line 108

3. Explanation

(1) Root Cause

The polling condition checked `media.status !== MEDIA_STATUS.PROCESSING` instead of checking whether the media was complete.

When status was COMPLETE, it still satisfied that condition and returned 202 Accepted, so the download never redirected.

(2) Fix

Changed the condition to `media.status !== MEDIA_STATUS.COMPLETE`, so only unfinished media returns 202 and completed media redirects.

4. Code Diff

```
message: media processing in progress. hummingbird-production $ git diff controllers/media.js
diff --git a/controllers/media.js b/controllers/media.js
index 6bb1c50..8f3b61f 100644
--- a/controllers/media.js
+++ b/controllers/media.js
@@ -105,10 +105,10 @@ const downloadController = async (req, res) => {
   return;
 }

- if (media.status !== MEDIA_STATUS.PROCESSING) {
+ if (media.status !== MEDIA_STATUS.COMPLETE) {
   const SIXTY_SECONDS = 60;
   res.set('Retry-After', SIXTY_SECONDS);
-   res.set('Location', `${req.hostname}/v1/media/${mediaId}/status`);
+   res.set('Location', `http://${req.get('host')}/v1/media/${mediaId}/status`);
   logger.info(
     { mediaId, currentStatus: media.status },
     'Media not ready for download, sending 202'
   );
   return;
 }
```

5. Verify the Fix Works

```
humming-bird-midterm $ curl -X POST "$ALB/v1/media/upload?width=500" -F "file=@sample.jpg"
{"mediaId":"d50deac7-3374-44e6-8d14-04e1ed216720"}humming-bird-midterm $ export MID=d50deac7-3374-44e6-8d14-04e1ed216720
humming-bird-midterm $ curl -X PUT "$ALB/v1/media/$MID/resize?width=500"
{"mediaId":"d50deac7-3374-44e6-8d14-04e1ed216720","status":"COMPLETE"}humming-bird-midterm $ curl "$ALB/v1/media/$MID/status"
{"status":"COMPLETE"}humming-bird-midterm $ curl -i "$ALB/v1/media/$MID/download"
HTTP/1.1 302 Found
Date: Mon, 09 Mar 2026 07:53:07 GMT
Content-Type: text/plain; charset=utf-8
Content-Length: 1636
Connection: keep-alive
X-Powered-By: Express
Location: https://hummingbird-production-409684939360.s3.us-west-2.amazonaws.com/uploads/d50deac7-3374-44e6-8d14-04e1ed216720/sample.jpg?
y%2BVao4N6wIhAK3n95XpKyhEhDrLnYQRLRZVMD8s24CwRY5uKnuRKkV8DCCKQABoMNDASNjg00TM5MzYwIgzRPTWhm0uYWS0eFrMq3AMWEyMaVx2JawUSpGaVrsJmiPprn%2B4
DWMVP73sspZb6T8xgwphsLCznjynB9QkrJczBiu%2FgUdIiIW7WRcXgW%2FYt5IT80qbhwvLiIJu%2FFA4bMcAdE%2BL6SYuN1v%2FMvbElnn4BgK%2BqF%2B18%2BZcVJu5xg%
FN1aYh10SaB0S%2FI%2BL71qZxgI0gaCVi9kUt39k0ja3jCs0roJaqSmALZeJZwZP%2B5Ue1Xny7uNUUnX258hqTUKNRliGkikNjM%2FFa7ZX%2B%2BTDZwu98p16YnEDYWB1VmQb1
Vary: Accept

Found. Redirecting to https://hummingbird-production-409684939360.s3.us-west-2.amazonaws.com/uploads/d50deac7-3374-44e6-8d14-04e1ed216720
SJKiXLIIVEFdCy%2BVao4N6wIhAK3n95XpKyhEhDrLnYQRLRZVMD8s24CwRY5uKnuRKkV8DCCKQABoMNDASNjg00TM5MzYwIgzRPTWhm0uYWS0eFrMq3AMWEyMaVx2JawUSpGaV
RxSPQRsN4GPlDWMVP73sspZb6T8xgwphsLCznjynB9QkrJczBiu%2FgUdIiIW7WRcXgW%2FYt5IT80qbhwvLiIJu%2FFA4bMcAdE%2BL6SYuN1v%2FMvbElnn4BgK%2BqF%2B18
8sptqfn0dcxoFN1aYh10SaB0S%2FI%2BL71qZxgI0gaCVi9kUt39k0ja3jCs0roJaqSmALZeJZwZP%2B5Ue1Xny7uNUUnX258hqTUKNRliGkikNjM%2FFa7ZX%2B%2BTDZwu98p16Y
```

Bonus

1. Reproduce the Bug

```
{ "status": "PENDING" } humming-bird-midterm $ curl -X POST "$ALB/v1/media/upload?width=500" -F "file=@sample.jpg"
{ "mediaId": "2aff27bc-e5e2-4303-812b-686bfffac28a2" } humming-bird-midterm $ export MID=2aff27bc-e5e2-4303-812b-686bfffac28a2
humming-bird-midterm $ curl -X PUT "$ALB/v1/media/$MID/resize?width=500"
{ "mediaId": "2aff27bc-e5e2-4303-812b-686bfffac28a2", "status": "COMPLETE" } humming-bird-midterm $ curl "$ALB/v1/media/$MID/status"
{ "status": "PENDING" } humming-bird-midterm $ curl "$ALB/v1/media/$MID/status"
{ "status": "PENDING" } humming-bird-midterm $ curl "$ALB/v1/media/$MID/status"
{ "status": "PENDING" } humming-bird-midterm $
```

2. Bug Location

clients/dynamodb.js - setMediaStatus

line 155

3. Explanation

(1) Root Cause

setMediaStatus used SK: 'metadata' while createMedia and getMedia used SK: 'METADATA'.

DynamoDB keys are case-sensitive, so the update targeted a different item. Because UpdateItem had no ConditionExpression, it silently created a new row instead of updating the original metadata row.

(2) Fix

changed the sort key to SK: { S: 'METADATA' } so all three functions target the same item.

4. Code Diff

```
diff --git a/clients/dynamodb.js b/clients/dynamodb.js
index 700494c..4291536 100644
--- a/clients/dynamodb.js
+++ b/clients/dynamodb.js
@@ -81,6 +81,7 @@ const getMedia = async (mediaId) => {
    name: Item.name.S,
    mimetype: Item.mimetype.S,
    status: Item.status.S,
+   width: Number(Item.width.N),
  };
} catch (error) {
  logger.error(error);
@@ -151,7 +152,7 @@ const setMediaStatus = async ({ mediaId, newStatus }) => {
  TableName,
  Key: {
    PK: { S: `MEDIA#${mediaId}` },
-   SK: { S: 'metadata' },
+   SK: { S: 'METADATA' },
  },
  UpdateExpression: 'SET #status = :newStatus',
  ExpressionAttributeNames: { '#status': 'status' },
```

5. Verify the Fix Works

```
humming-bird-midterm $ curl -X POST "$ALB/v1/media/upload?width=500" -F "file=@sample.jpg"
{"mediaId":"aa95a3e2-0e22-495b-9d06-65bd7e3c6fda"}humming-bird-midterm $ export MID=aa95a3e2-0e22-495b-9d06-65bd7e3c6fda
humming-bird-midterm $ curl -X PUT "$ALB/v1/media/$MID/resize?width=500"
{"mediaId":"aa95a3e2-0e22-495b-9d06-65bd7e3c6fda","status":"COMPLETE"}humming-bird-midterm $ curl "$ALB/v1/media/$MID/status"
{"status":"COMPLETE"}humming-bird-midterm $ curl "$ALB/v1/media/$MID/status"
{"status":"COMPLETE"}humming-bird-midterm $ curl "$ALB/v1/media/$MID/status"
{"status":"COMPLETE"}humming-bird-midterm $ █
```